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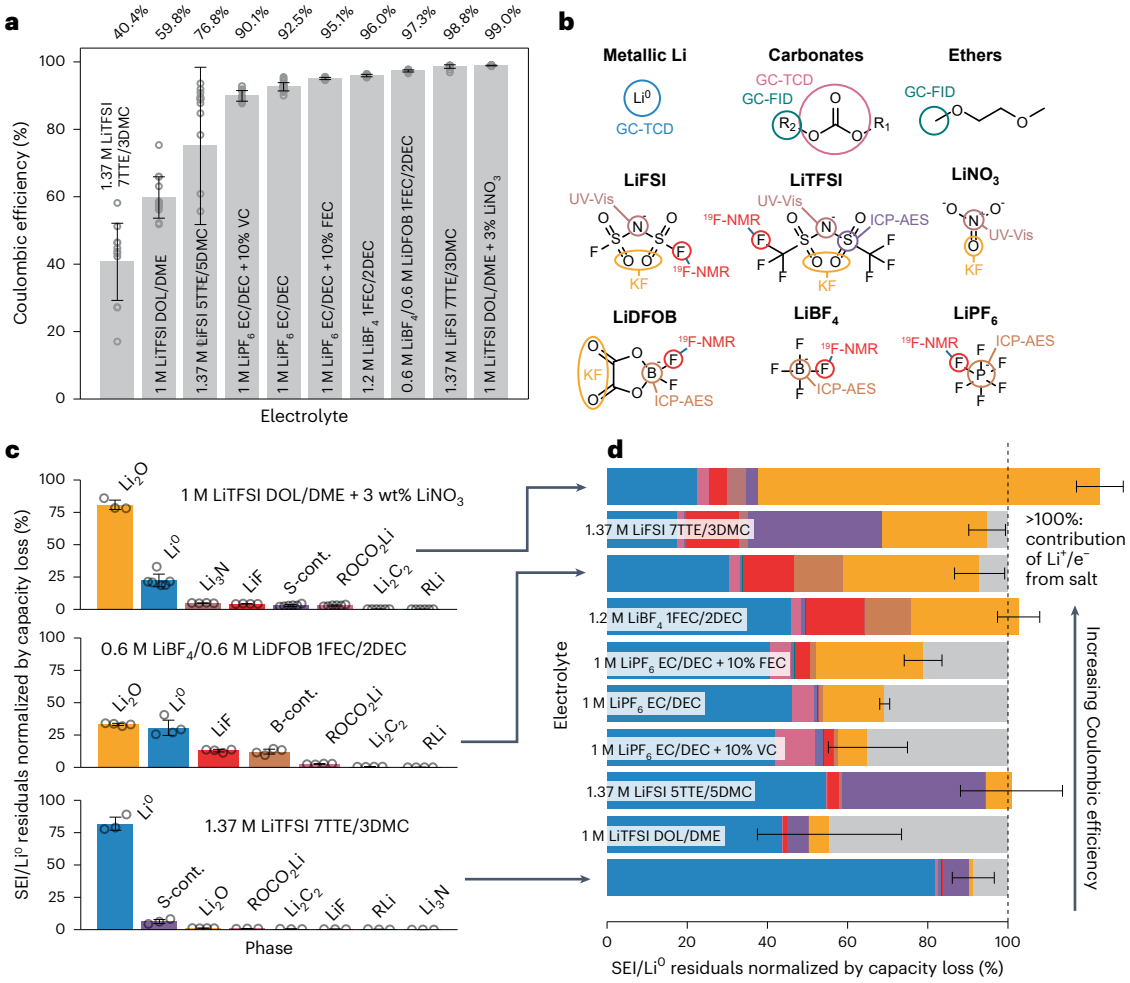


**Fig. 2 | Quantification of SEI/Li$^0$ residuals in diverse electrolytes. a**, Coulombic efficiency across diverse electrolytes, determined by the same cycling protocol used to generate the titration samples (Supplementary Note 4). Data presented as mean ± standard deviation, and open markers denote individual cells ($n$ = 10, 13, 11, 11, 12, 12, 12, 12, 15, 13 for electrolytes in order of increasing CE). Lower-CE electrolytes consistently displayed larger scatter than high-CE electrolytes. **b**, Examples of solvent species and salts present in the electrolytes considered herein, annotated with the instrument/titration techniques used to measure decomposition products from the indicated chemical moieties. Techniques include gas chromatography with thermal conductivity (GC-TCD) and flame ionization (GC-FID) detectors, $^{19}$F nuclear magnetic resonance spectroscopy (NMR), ultraviolet-visible spectroscopy (UV-Vis), ICP-AES, and KF titration. **c**, Measured SEI/Li$^0$ residuals normalized by capacity loss for select electrolytes in the series. Data are presented as mean ± standard deviation. Open markers denote individual data points (top frame: $n$ = 6 for Li$^0$, S, $ROCO_2Li$, $Li_2C_2$, RLi; $n$ = 4 for $Li_3N$, LiF; $n$ = 3 for $Li_2O$; middle frame: $n$ = 4 for all phases; bottom frame: $n$ = 3 for all phases). **d**, Cumulative representation of the average of each SEI/Li$^0$ residual normalized by capacity loss for all electrolytes in the series, organized by CE. Colours as in **c**. The error bars denote the estimated cumulative standard deviation of all phases, calculated from the root square sum of the error bars in **c** and centred around the sum of the mean of all phases. Sample number '$n$' for each electrolyte equal to **a**. Raw data for each electrolyte can be found in Source Data and Supplementary Figs. 15–24. In electrolytes for which there is extensive anion decomposition, >100% cumulative SEI/Li$^0$ residuals can occur due to contributions from charged salt ions to the Li inventory on the cycled electrode (Supplementary Note 6 for $LiNO_3$ as an example). Supplementary Fig. 25 shows the data for the SEI phases in **d** but normalized by SEI, rather than total, capacity loss (that is, total capacity loss minus inactive Li$^0$).

The next electrolytes comprise a series of conventional carbonates: 1 M $LiPF_6$ EC/DEC + 10 vol% vinylene carbonate (VC) (CE = 90.1%); and the same without additive (CE = 92.5%) or with 10 vol% fluoroethylene carbonate (FEC) (CE = 95.1%). All have marked presence of $ROCO_2Li$, as expected[40]. However, increasing CE corresponds to increasing preponderance of $Li_2O$, from being nearly comparable to $ROCO_2Li$ for the VC-based electrolyte (7.0% of total loss) to well-exceeding $ROCO_2Li$ with FEC (26.4%). With fluorinated $LiPF_6$, $Li_2O$ can only originate from solvent-derived phases such as $ROCO_2Li$[41] and oligo/polycarbonates from VC or FEC[42]. Reasonably, LiF content was highest with FEC.

Electrolytes with higher CE (generally >90%) exhibit SEI increasingly dominated by $Li_2O$. In 1.2 M $LiBF_4$ and 0.6 M $LiBF_4$/0.6 M lithium difluoro(oxalato)borate (LiDFOB) in 1FEC/2DEC (CE = 96.0% and 97.3%, respectively), $Li_2O$ is followed by notable contributions from B-containing (B-cont.) phases (Supplementary Figs. 32 and 33) and LiF resulting from salt and FEC breakdown. Meanwhile, the LHCE 1.37 M LiFSI 7TTE/3DMC (CE = 98.8%) has a more chemically diverse SEI[38] with major contributions from S-containing phases (for example, $-SO_2F$; Supplementary Fig. 34), followed by $Li_2O$/LiF and more minor contributions of $ROCO_2Li$ and $Li_3N$.

Finally, an electrolyte containing $LiNO_3$, an additive known to promote $Li_2O$ formation[28,43], was examined (1 M LiTFSI DOL/DME + 3 wt% $LiNO_3$, CE = 99.0%). This electrolyte yielded SEI capacity losses overwhelmingly dominated by $Li_2O$ (80.8%), indicating that the relationship between $Li_2O$ and CE, examined more rigorously below, can be nonlinear. LiF, which in this electrolyte can only originate from C–F in LiTFSI, was proportionally much lower (4.3%). The same electrolyte was also tested with only 1% $LiNO_3$ and yielded similar CE and $Li_2O$ (Supplementary Fig. 36). Notably, this electrolyte's cumulative quantification exceeded 100% of capacity loss when normalized to measured charge.

**Table 1 | Summary of CEs and ionic phase composition of analysed electrolytes**

| Electrolyte | CE (%) | $Li_2O$ (%)[1] | LiF (%)[1] |
|---|---|---|---|
| 1.37 M LiTFSI 7TTE/3DMC (70/30 vol%) | 40.4 | 1.1 | 0.4 |
| 1 M LiTFSI DOL/DME (50/50 vol%) | 59.8 | 4.9 | 0.8 |
| 1.37 M LiFSI 5TTE/5DMC (50/50 vol%) | 76.8 | 4.4 | 2.6 |
| 1 M $LiPF_6$ EC/DEC (50/50 vol%) + 10 vol% VC | 90.1 | 7.0 | 2.4 |
| 1 M $LiPF_6$ EC/DEC (50/50 vol%) | 92.5 | 14.6 | 0.2 |
| 1 M $LiPF_6$ EC/DEC (50/50 vol%) + 10 vol% FEC | 95.1 | 26.4 | 3.8 |
| 1.2 M $LiBF_4$ 1FEC/2DEC (33/66 vol%) | 96.0 | 26.1 | 14.5 |
| 0.6 M $LiBF_4$ 0.6 M LiDFOB 1FEC/2DEC (33/66 vol%) | 97.3 | 33.2 | 12.9 |
| 1.37 M LiFSI 7TTE/3DMC (70/30 vol%) | 98.8 | 24.9 | 13.1 |
| 1 M LiTFSI DOL/DME (50/50 vol%) + 3 wt% $LiNO_3$ | 99.0 | 80.8 | 4.3 |

[1]Indicated $Li_2O$ and LiF amounts are normalized to the irreversible capacity loss. Information on other phases can be found in Supplementary Table 9. Electrolyte nomenclatures are defined in text and Supplementary Methods.

This scenario indicates an SEI-forming contribution from the salt, here $Li^+ NO_3^-$, that can form $Li_2O$ without electrons being counted by the potentiostat (Supplementary Note 6). Across all electrolytes with N-containing salts (LiTFSI, LiFSI, $LiNO_3$), this electrolyte exhibited the highest amount of fully reduced $Li_3N$ (4.8%). Together, the above results provide substantially improved resolution to SEI composition, enabling further scrutiny of compositional correlations with CE.

## Statistical correlations among SEI phases and CE

The relationship of a given phase with CE is generally not monotonic (Supplementary Fig. 37). To rigorously identify statistical correlations within the dataset, the Spearman rank correlation coefficient ($\rho$) was utilized (Supplementary Note 7)[44]. Electrolytes were ranked in order of increasing CE and in proportion of each phase (%), and the linear coefficient of correlation between ranks, $\rho$, was calculated (Fig. 3a and Supplementary Fig. 38), with $\rho = 1/-1/0$ for strictly positive/negative/neutral correlations. Associated with each $\rho$, a degree of statistical significance $\sigma$ quantifies the null hypothesis probability that an observed correlation could have originated by chance from uncorrelated variables ($\sigma > 2.5$ corresponds to <1%; Fig. 3b). Confidence intervals around each $\rho$ were also calculated to account for the effect of potential outliers in the dataset (Fig. 3c).

Expectedly[7], a clear inverse relationship between CE and inactive $Li^0$ was confirmed (Fig. 3a,b, $\rho = -0.818$, $\sigma = 2.77$), and LiF exhibited a positive albeit weaker correlation with CE ($\rho = 0.758$, $\sigma = 2.42$). However, $Li_2O$ showed the strongest and most statistically significant correlation across all capacity loss contributions ($\rho = 0.903$, $\sigma = 3.29$), notably stronger than the well-established negative relationship with $Li^0$ (ref. 7). Furthermore, when phases are normalized to SEI loss only, rather than total capacity loss (Supplementary Fig. 25), $Li_2O$ remains the most correlated with CE, whereas the positive correlation of LiF became less significant (Supplementary Figs. 39 and 40). All other SEI phases showed weaker statistical significance for CE ($\sigma < 1$; Fig. 3b). However, it is emphasized that trends in $Li_3N$, P-, S- and B-content could not be as rigorously examined because the electrolytes considered do not uniformly span all elemental diversity equally, leading to wide confidence intervals in $\rho$ for these phases (Fig. 3c). Regardless, excepting S-containing phases, these were not typically seen to be major SEI components, and none (including S) was required to achieve high CE, unlike $Li_2O$. Similarly, the remaining solvent-derived phases RLi, $ROCO_2Li$ and $Li_2C_2$ showed weak correlation with CE and wide confidence intervals.

To further test the relationship between $Li_2O$ and CE, $Li_2O$ was measured for four additional lower-CE and four other >99% CE electrolytes selected from Supplementary Fig. 1[45–48]. Fig. 3d shows the CE vs $Li_2O$ relationship for each cycled cell (in grey) and, in yellow, the $Li_2O$ and CE per electrolyte, averaged over at least three cells per electrolyte. The rank correlation over all cells (Fig. 3e) confirms the high coefficient of correlation between CE and $Li_2O$ ($\rho = 0.88$) but now with even stronger statistical significance ($\sigma = 3.7$).

## $Li_2O$ microstructure within the SEI

Titration-based correlations capture the ensemble composition of the SEI, but deciphering the role of $Li_2O$ in improving CE requires closer scrutiny of its distribution and function. Cryo-HRTEM was leveraged to localize crystalline phases in low- (1.37 M LiTFSI 7TTE/3DMC) and high-CE (1.37 M LiFSI 7TTE/3DMC, 1 M LiTFSI DOL/DME + 3 wt% $LiNO_3$) SEIs from the prior dataset (Fig. 4). These experiments were first conducted at low capacity (0.1 mAh $cm^{-2}$) to ensure samples remained electron transparent. Varied Li morphology was observed, from needle-like to hexagonally faceted to oblate, respectively, in order of increasing CE (Fig. 4a). The faceted deposition morphology is evidence of unimpeded $Li^+$ diffusion through a thin and/or permissive SEI[49] and is consistent with the lowest SEI thickness (7 nm) in the LiFSI 7TTE/3DMC electrolyte (Fig. 4b). On the other hand, the highest-CE electrolyte exhibited the thickest SEI (27 nm), which is consistent with the high degree of $LiNO_3$ reactivity observed in Fig. 2d and indicates that the relationship between CE and SEI thickness need not be strictly inverse. Further characterization was performed by selective area electron diffraction (SAED) and energy dispersive spectroscopy (EDS). SAED revealed diffraction rings attributed to $Li_2O$ exclusively, and notably no LiF reflections were found in any electrolyte at this plated capacity (Fig. 4c). However, F-, N- and S-containing phases were detected on all samples by EDS (Supplementary Fig. 41), consistent with titration (Fig. 2), thus indicating that the chemically diverse phases seen by these techniques are not crystalline at this plated capacity. HRTEM revealed $Li_2O$ particles in all samples but with notably distinct spatial distribution between the low- and high-CE SEIs (Fig. 4d). In the two high-CE electrolytes, a uniform ~5–10 nm $Li_2O$ layer was found oriented parallel to the SEI surface. Between the outer $Li_2O$ layer and the metallic Li core, larger, randomly oriented crystalline lattices were also found within an amorphous SEI matrix for the DOL/DME electrolyte, which agrees with the very high $Li_2O$ content (Fig. 2d). In the low-CE electrolyte, on the other hand, a primarily mosaic-like SEI nanostructure[50] was observed, with few $Li_2O$ particles dispersed discontinuously at the outer SEI region. Thus, the benefits imparted to the SEI by $Li_2O$ content are most strongly associated with a continuous localization facing the electrolyte[51], with the additional mosaic-like inclusions in the DOL/DME SEI boosting $Li_2O$ content but not substantially improving CE.

To better understand the apparent absence of crystallized LiF, further imaging was conducted on the 1.37 M LiFSI 7TTE/3DMC LHCE electrolyte. At a higher capacity (0.2 mAh $cm^{-2}$), high aspect ratio deposits were now observed (Fig. 4e), and the SEI showed evidence of emerging crystalline LiF superposed with $Li_2O$ (Fig. 4f–h). This delayed onset of LiF formation during Li deposition indicates that phases containing F, S and N from the $FSI^-$ anion remain amorphous in the SEI matrix and are only later reduced and crystallized, possibly due to contact with subsequently plated $Li^0$. These findings were further verified by $^{19}$F-NMR, which showed a greater abundance of $-SO_2F$ at low capacity relative to LiF, the amounts of which inverted once more Li was plated (Fig. 4i and Supplementary Fig. 42–44). Thus, whereas LiF is a product of SEI evolution during Li plating, the initial SEI and morphology of Li are governed primarily by organofluorine and readily formed $Li_2O$.

## $Li_2O$ formation in localized high-concentration electrolytes

To further interrogate the role of $Li_2O$, an electrolyte series based on LiFSI 7TTE/3DMC was examined as a function of salt concentration, spanning below (0.25–1.25 M) and up to (1.37 M) the LHCE threshold.

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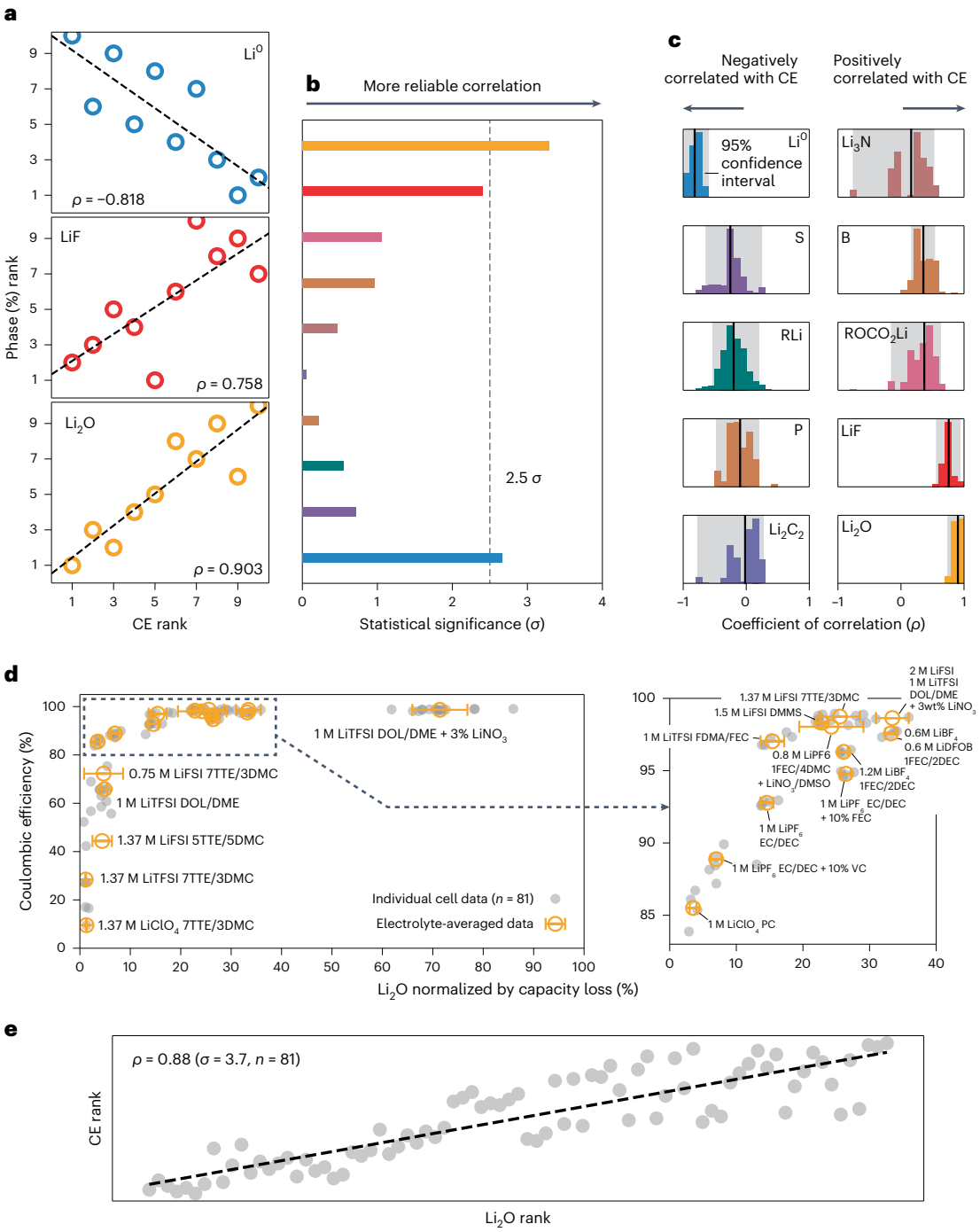


**Fig. 3 | Statistical correlations between quantifiable SEI/Li⁰ residual phases and CE. a**, Rank CE and rank phase (%) for inactive $Li^0$, LiF and $Li_2O$. The coefficient of correlation $\rho$ measures whether the relationship between CE and a given phase is monotonic, with $\rho = 1$ perfectly monotonic and positively correlated and $\rho = -1$ perfectly monotonic and negatively correlated. Each data point indicates a different electrolyte. **b**, Statistical significance $\sigma$ associated with each phase-CE correlation. **c**, Frequency histograms of $\rho$, calculated from all possible combinations of (ten choose seven) data points (that is, excluding up to three data points from the dataset). The solid vertical line indicates $\rho$ calculated from all ten data points, and the shaded region indicates the 95% confidence interval calculated from the frequency histogram. **d**, CE vs $Li_2O$ relationship, now including eight additional electrolytes (18 total; additional electrolytes are 1.37 M $LiClO_4$ 7TTE/3DMC, 0.5 M and 0.75 M LiFSI 7TTE/3DMC, 1 M $LiClO_4$ propylene carbonate (PC), 1.5 M LiFSI dimethoxydimethylsilane (DMMS), 1 M LiTFSI 2,2,2-trifluoro-N,N-dimethylacetamide (FDMA)/FEC, 0.8 M $LiPF_6$ 1FEC/3DMC + 5 wt% 4 M $LiNO_3$/dimethyl sulfoxide (DMSO) and 2 M LiFSI/1 M LiTFSI DOL/DME + 3 wt% $LiNO_3$), for a total of $n = 81$ samples. CE and $Li_2O$ were measured using cycling conditions that meet the capacity loss requirements for 2-butoxyethanol titration. Error bars indicate the standard deviation of each electrolyte, over at least three cells. **e**, Rank correlation from **d**.

LiFSI is the most widely used salt used in high-CE electrolytes (Supplementary Fig. 1), where the higher salt concentrations enhance $Li^+$-$FSI^-$ pairing, confirmed here via diffusion-ordered spectroscopy (DOSY)-NMR (Fig. 5a), that promote anion-derived SEI important for high CE[38,52,53]. The first-cycle CEs (4 mAh $cm^{-2}$, 0.5 mA $cm^{-2}$) are shown in Fig. 5b.

Capacity loss breakdowns found $Li^0$ to be the major contributor at lower CE (16.5–91.4%, 0.25–0.75 M; Supplementary Fig. 45). From 0.75 M